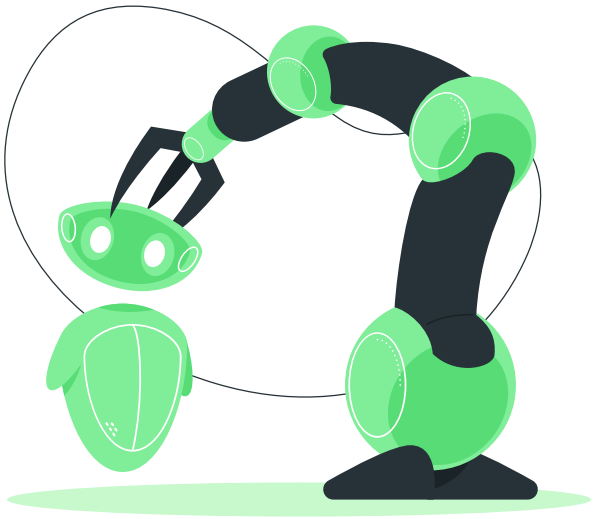
Robotics
CS 3015

Umarah Qaseem

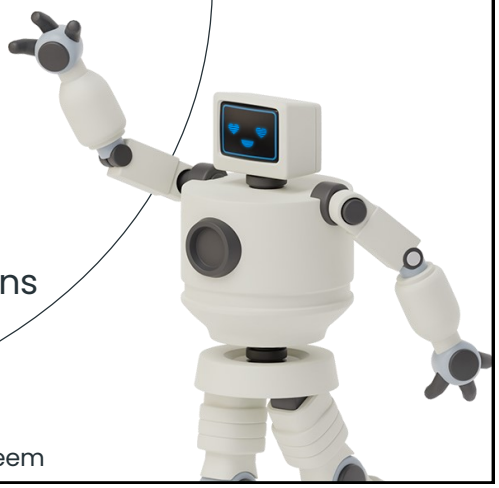


1

8

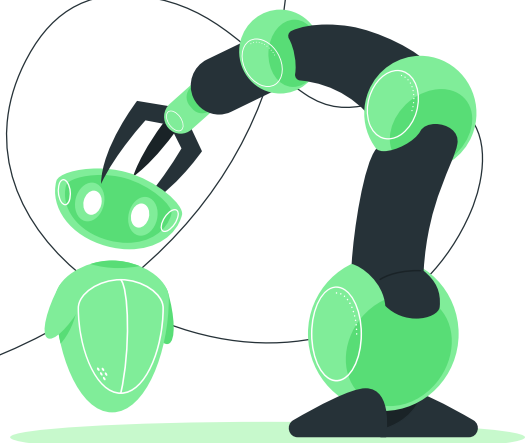
- Robot Motion
- Joint Interpolated Motion
- A Smooth Trajectory
- Joint Trajectory
- Cartesian Trajectory

- Path and Trajectories
- Quintic Polynomial and its derivatives, Boundary conditions
- Multi-segment Trajectories
- Blending the edges



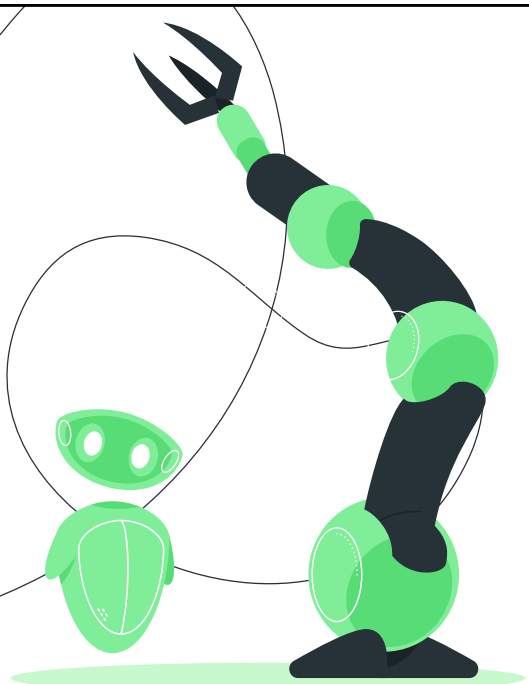
2

Robot Motion



4

Joint Interpolated Motion



5

Joint Interpolated motion

- Goal: Move smoothly from one pose to another.
- Suppose we want to move from pose A to pose B

$$\xi_A \rightarrow \xi_B$$

- We can determine the joint angles for each pose

$$q_A = \mathcal{K}^{-1}(\xi_A)$$
$$q_B = \mathcal{K}^{-1}(\xi_B)$$

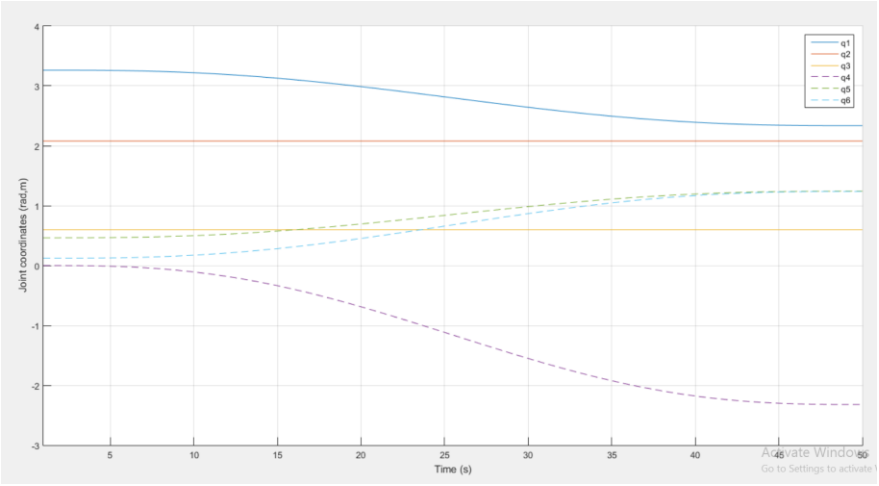
- The goal is to move smoothly $q_A \rightarrow q_B$

Robotics – Umarah Qaseem

6

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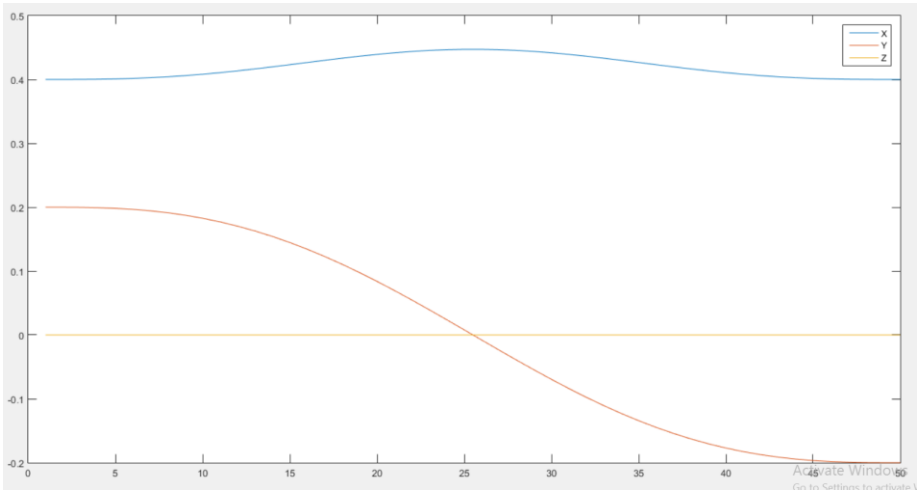
Joint Interpolated Motion - Analysis



Joint Angle variation over time

7

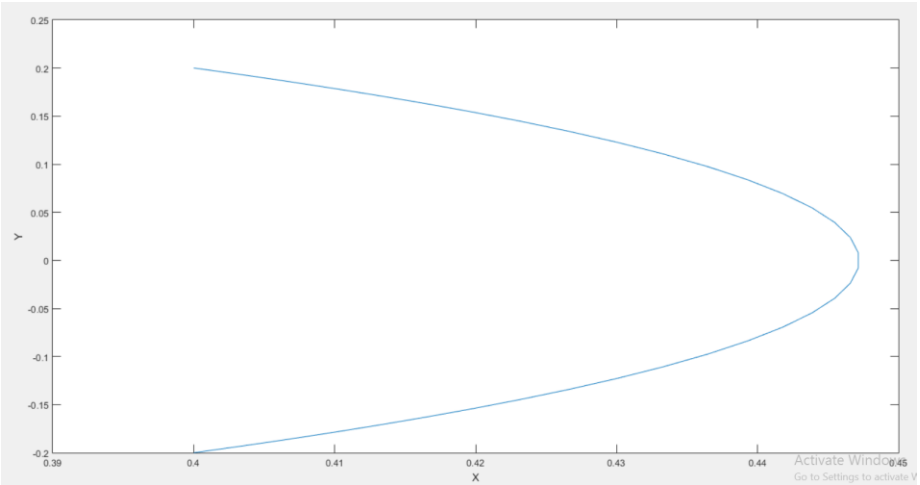
Joint Interpolated Motion – Analysis



Cartesian motion of end effector over time

8

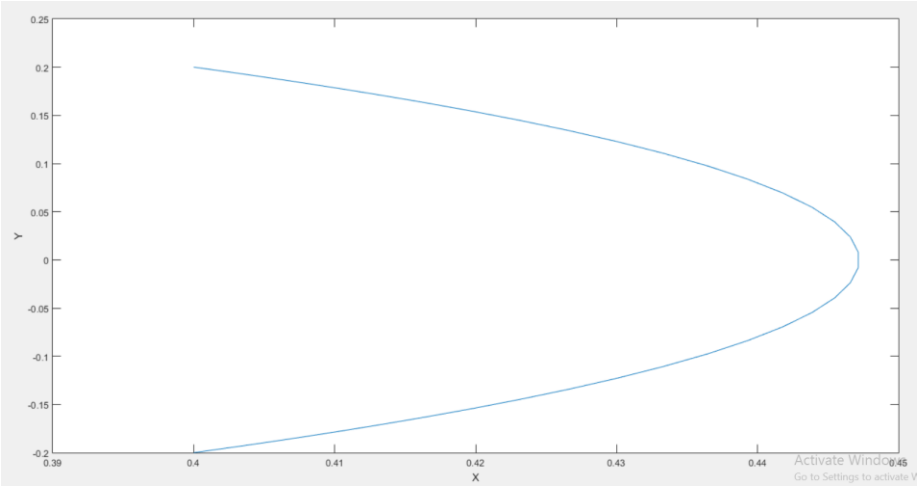
Joint Interpolated Motion – Analysis



Cartesian motion of end effector over time in XY Plane

9

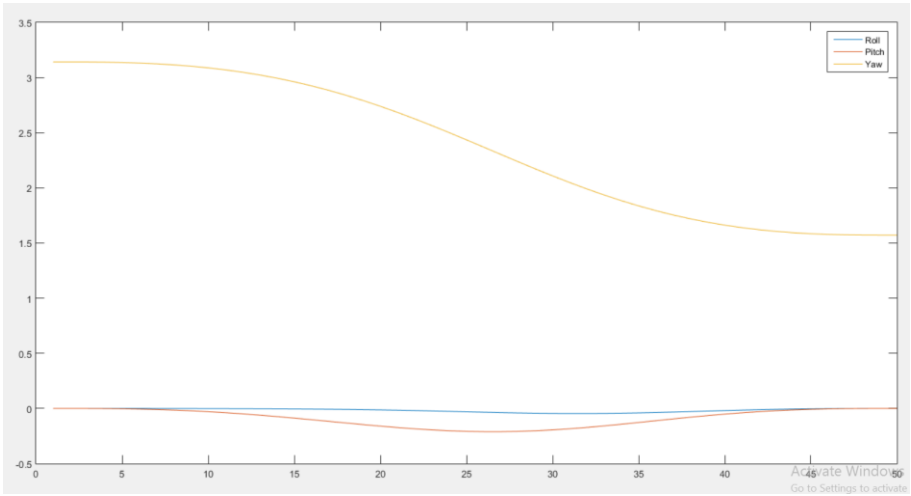
Joint Interpolated Motion – Analysis



Cartesian motion of end effector over time in XY Plane

10

Joint Interpolated Motion – Analysis



RPY variation over time

11

Cartesian Interpolated Motion

- **Goal**: we want to smoothly move from pose A to pose B
- We interpolate pose instead of joint angles

$$\xi_A \rightarrow \xi_B$$

- Then to get joint angles we simply apply inverse kinematics

$$\xi_s = f(\xi_A, \xi_B, s),\ 0 \leq s \leq 1$$

$$\boldsymbol{q}_s = \mathcal{K}^{-1}(\xi_s)$$

12

Cartesian Interpolated Motion

```
Ts=ctraj(TA, TB, 50);

qs = p560.ikine6s(Ts);

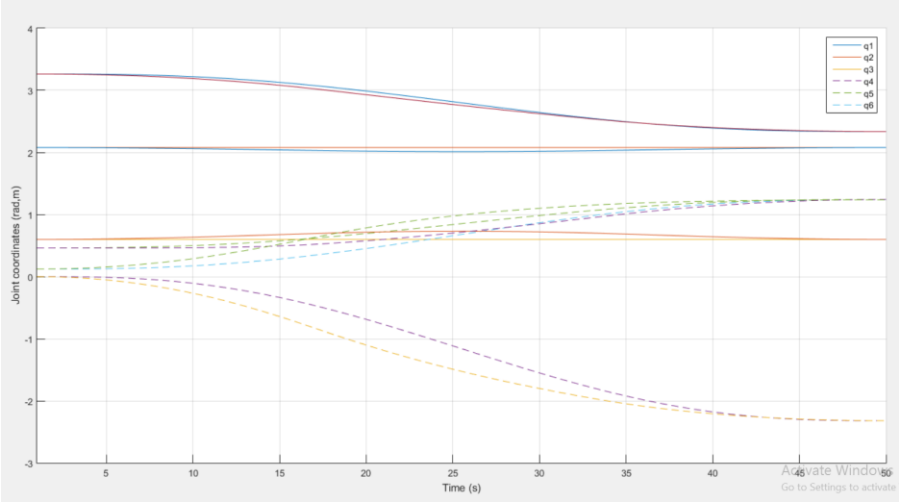
qplot(qs)

fk= p560.fkine(qs);
```

13

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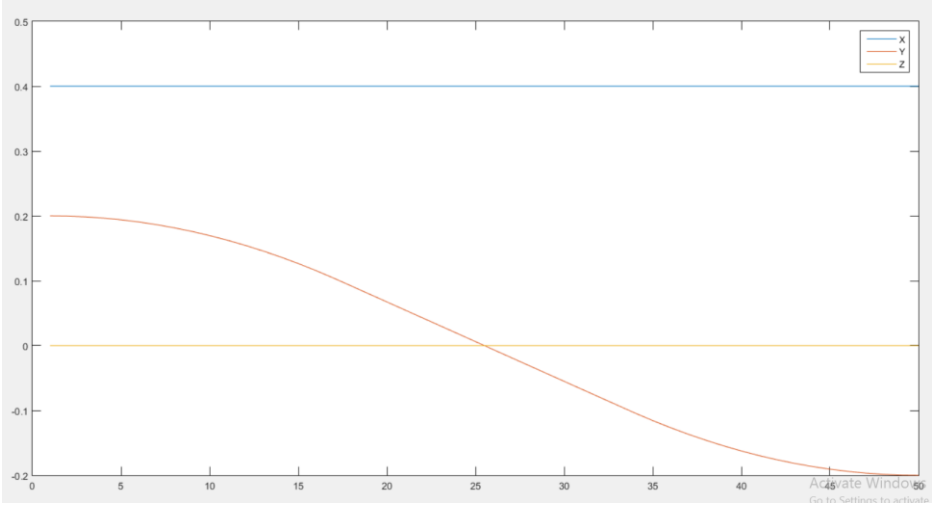
Cartesian Interpolated Motion



14

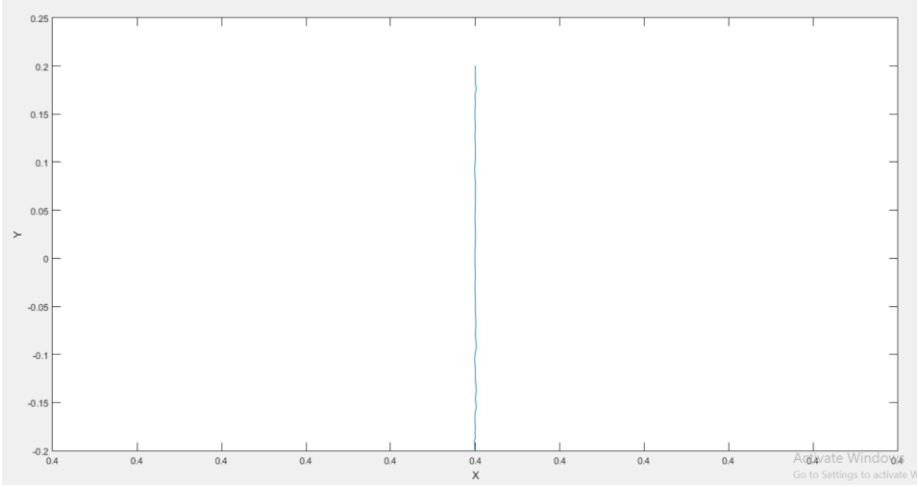
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Cartesian Interpolated Motion



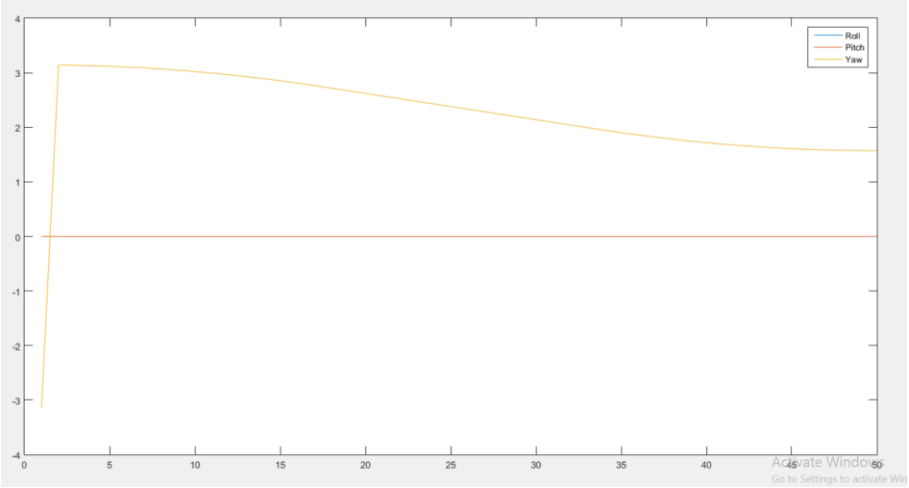
15

Cartesian Interpolated Motion



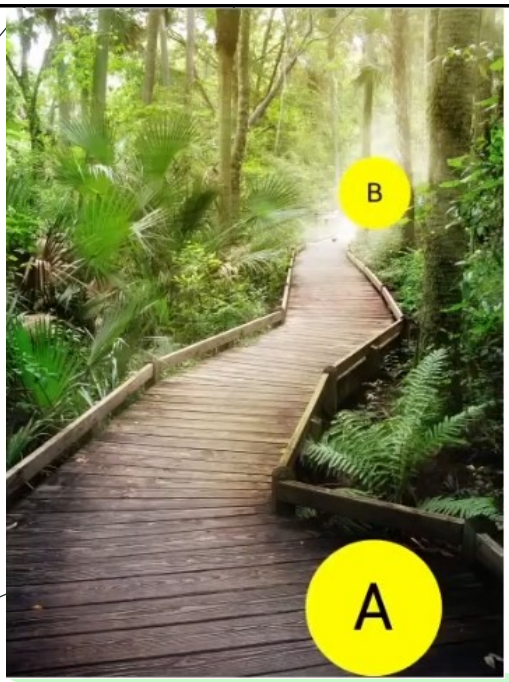
16

Cartesian Interpolated Motion



17

Path and Trajectories



18

Inertial Reference Frame

- Term Frequently used in Robotics
- **Definition**: *a reference frame that is not accelerating or rotating*
- If a frame {B} is moving at a constant velocity with respect to stationary reference frame, then according to Newton's First Law *"that in the absence of an applied force a particle moves at a constant velocity"* {B} is also an inertial reference frame.
- If frame {B} is accelerating i.e. it violates Newton's First law then it is non-inertial reference frame.

19

Inertial Reference Frame

- In robotics the term **inertial frame** and **world coordinate frame** tend to be used loosely and interchangeably to indicate a frame fixed to some point on the Earth. This is to distinguish it from the body-frame attached to the robot or vehicle.

21

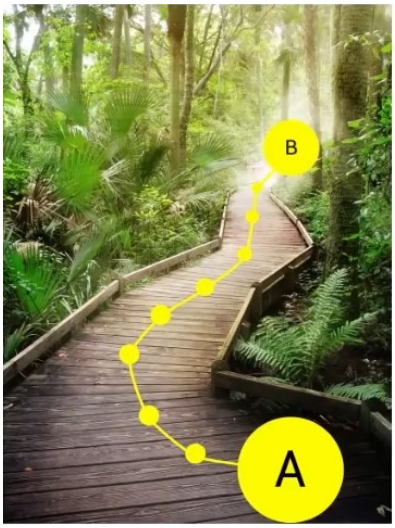
Creating Time-Varying Pose

- How pose changes as a function of time?
- Results in paths and trajectories
- We want the change in pose to be smooth

22

Path

- Way of getting from point A to B
- A step of points that link A to B
- No notion of speed that quantifies the progress along the path
  - No notion of time



23

Trajectories

- Path and schedule of getting from A to B
  - Notion of time

```
T0=eye(4)
T1=transl(1,2,3)*rpy2tr(0.6,0.8,1.4)

trplot(T0)
trplot(T1)
tranimate(T0,T1)
```

```
>> T0 = eye(4,4)

T0 =

     1     0     0     0
     0     1     0     0
     0     0     1     0
     0     0     0     1

>> T1 = transl(1,2,3)*rpy2tr(0.6, 0.8, 1.4)

T1 =

    0.1184   -0.6866    0.7174    1.0000
    0.8822   -0.2589   -0.3934    2.0000
    0.4558    0.6794    0.5750    3.0000
         0         0         0    1.0000
```

24

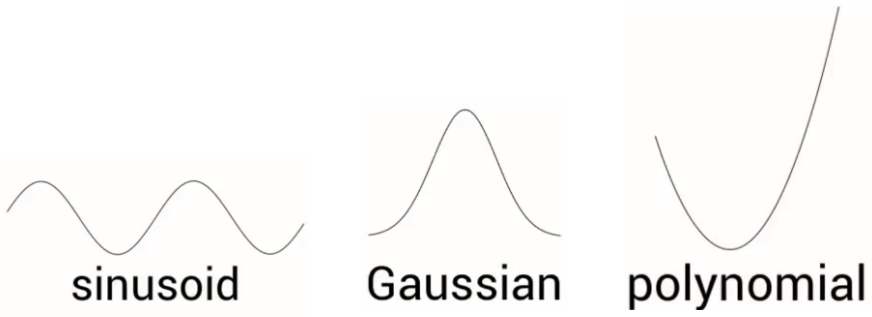
Smoothness

- Desired: Position and Orientation varies smoothly with time
  - Reduce peak acceleration of robot
  - Reduce the size of the motors required
  - Reduce vibrations in the structure of the robot
- Smoothness?
  - Position $x(t)$ is continuous
  - Velocity dx/dt is continuous
  - Acceleration d^2x/dt is continuous
  - Jerk is d^3x/dt continuous

25

Smooth Curve

- Variety of smooth mathematical functions



26

Polynomial

- Qintic (5th order) polynomial has 6 coefficients
- Thus, we can specify 6 boundary conditions
  - Position, velocity and acceleration at t=0 and t=T

$$s(t) = At^5 + Bt^4 + Ct^3 + Dt^2 + Et + F \qquad t \in [0, T]$$

time	s	$\dot{s}$	$\ddot{s}$
$t = 0$	s_0	$\dot{s}_0$	$\ddot{s}_0$
$t = T$	s_T	$\dot{s}_T$	$\ddot{s}_T$

27

Applying Boundary Conditions

$$s(t) = At^5 + Bt^4 + Ct^3 + Dt^2 + Et + F$$
$$\dot{s}(t) =$$
$$\ddot{s}(t) =$$

time	s	$\dot{s}$	$\ddot{s}$
$t = 0$	s_0	$\dot{s}_0$	$\ddot{s}_0$
$t = T$	s_T	$\dot{s}_T$	$\ddot{s}_T$

when $t = 0$

$$s(0) =$$
$$\dot{s}(0) =$$
$$\ddot{s}(0) =$$

when $t = T$

$$s(T) =$$
$$\dot{s}(T) =$$
$$\ddot{s}(T) =$$

28

Applying Boundary Conditions

$$s(t) = At^5 + Bt^4 + Ct^3 + Dt^2 + Et + F$$
$$\dot{s}(t) = 5At^4 + 4Bt^3 + 3Ct^2 + 2D + E$$
$$\ddot{s}(t) = 20At^3 + 12Bt^2 + 6Ct + 2D$$

time	s	$\dot{s}$	$\ddot{s}$
$t = 0$	s_0	$\dot{s}_0$	$\ddot{s}_0$
$t = T$	s_T	$\dot{s}_T$	$\ddot{s}_T$

when $t = 0$

$$s(0) = F$$
$$\dot{s}(0) = E$$
$$\ddot{s}(0) = 2D$$

when $t = T$

$$s(T) = AT^5 + BT^4 + CT^3 + DT^2 + ET + F$$
$$\dot{s}(T) = 5AT^4 + 4BT^3 + 3CT^2 + 2DT + E$$
$$\ddot{s}(T) = 20AT^3 + 12BT^2 + 6CT + 2D$$

29

Solve for Coefficients

- Writing in matrix form

$$\begin{pmatrix} s_0 \\ s_T \\ \dot{s}_0 \\ \dot{s}_T \\ \ddot{s}_0 \\ \ddot{s}_T \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ T^5 & T^4 & T^3 & T^2 & T & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 5T^4 & 4T^3 & 3T^2 & 2T & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 20T^3 & 12T^2 & 6T & 2 & 0 & 0 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \\ D \\ E \\ F \end{pmatrix}$$

30

Solve for Coefficients

- Invert the matrix to get coefficients

$$\begin{pmatrix} A \\ B \\ C \\ D \\ E \\ F \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ T^5 & T^4 & T^3 & T^2 & T & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 5T^4 & 4T^3 & 3T^2 & 2T & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 20T^3 & 12T^2 & 6T & 2 & 0 & 0 \end{pmatrix}^{-1} \begin{pmatrix} s_0 \\ s_T \\ \dot{s}_0 \\ \dot{s}_T \\ \ddot{s}_0 \\ \ddot{s}_T \end{pmatrix}$$

31

In terms of theta

$$\theta(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5$$

$$\begin{bmatrix} \theta_0 \\ \omega_0 \\ \alpha_0 \\ \theta_f \\ \omega_f \\ \alpha_f \end{bmatrix} = \begin{bmatrix} 1 & t_0 & t_0^2 & t_0^3 & t_0^4 & t_0^5 \\ 0 & 1 & 2t_0 & 3t_0^2 & 4t_0^3 & 5t_0^4 \\ 0 & 0 & 2 & 6t_0 & 12t_0^2 & 20t_0^3 \\ 1 & t_f & t_f^2 & t_f^3 & t_f^4 & t_f^5 \\ 0 & 1 & 2t_f & 3t_f^2 & 4t_f^3 & 5t_f^4 \\ 0 & 0 & 2 & 6t_f & 12t_f^2 & 20t_f^3 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix}$$

32

Toolbox

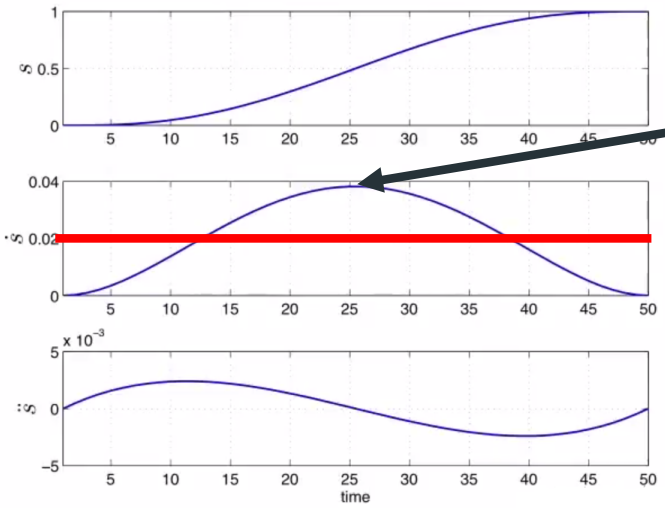
```
tpoly(0,1,50)                    % trajectory polynomial

S=tpoly(0,1,50)            % to get positions

[s sd sdd] = tpoly(0,1,50)      % to get position,  velocity and acceleration
```

35

Performance Problem (one limitation)



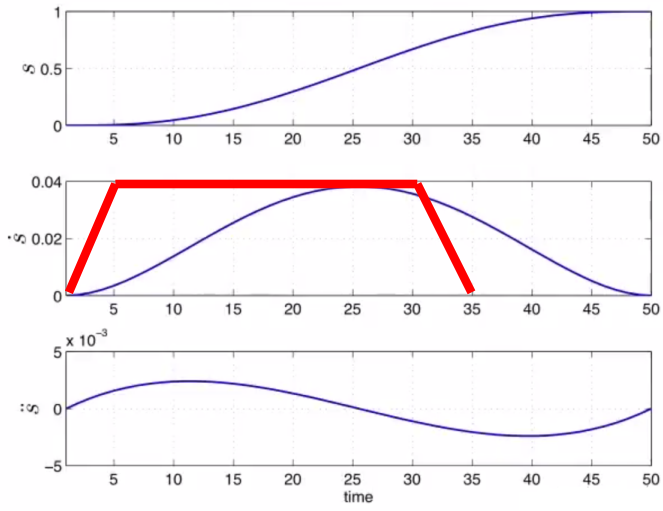
Peak velocity
0.038

Average velocity
0.02

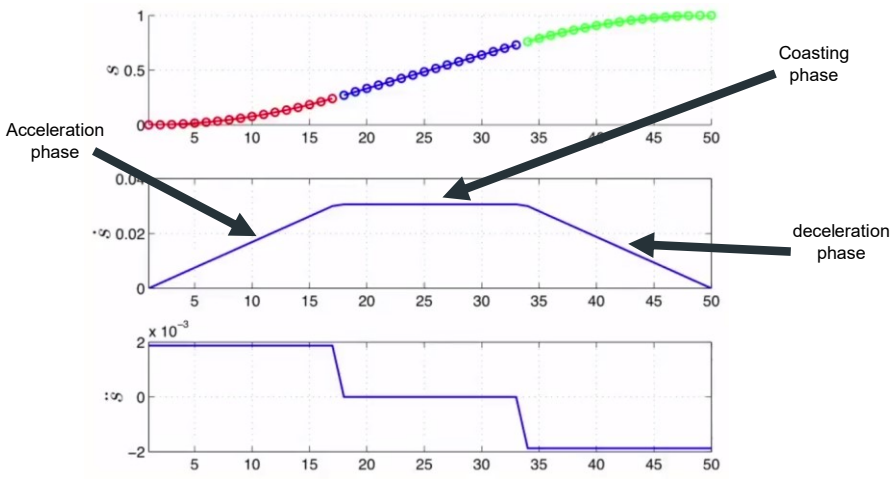
*Average velocity is only 52% of peak or the velocity at which we can move…..

36

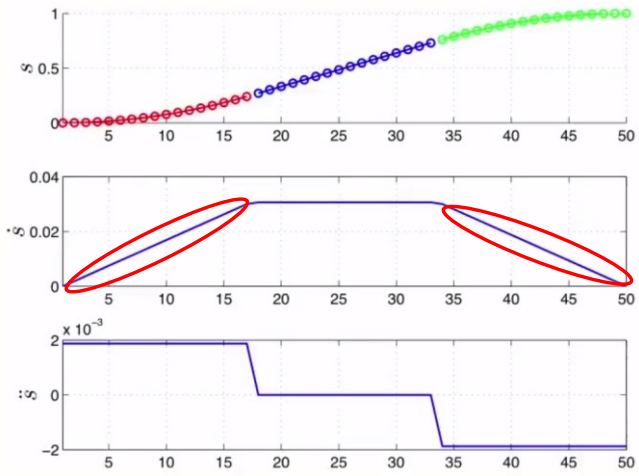
Alternative Velocity Profile



Tapezoidal Profile



Tapezoidal Profile



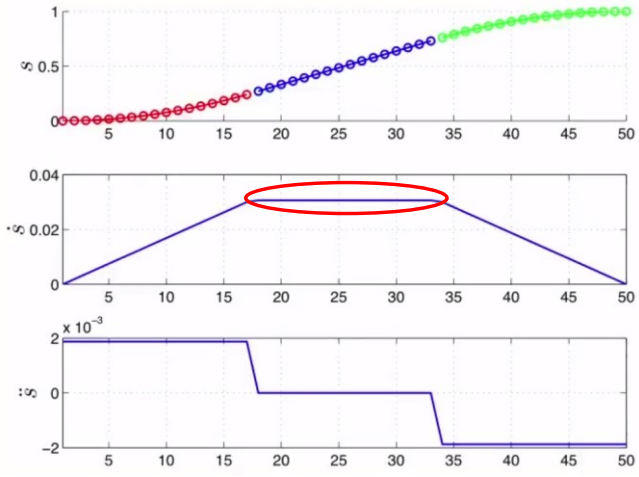
Acceleration/deceleration Phase

$$\dot{s} = at + b$$

$$s = at^2 + bt + c$$

parabola

Tapezoidal Profile



Coasting Phase

$$\dot{s} = b$$

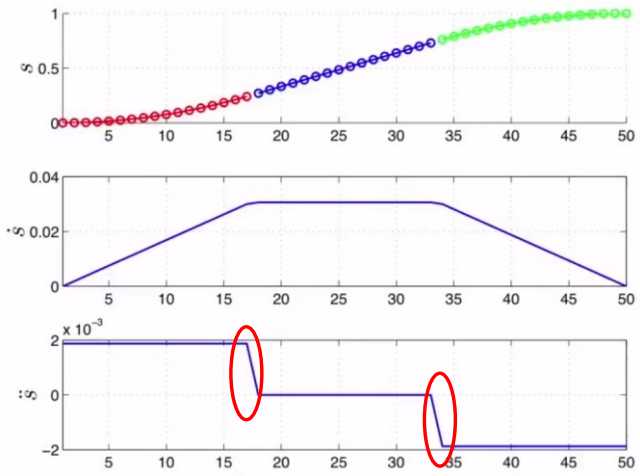
$$s = bt + c$$

straight line

A.K.A linear segment with parabolic blend

Toolbox: lspb

Problem



*Problematic for some robots, however generally we assume that bandwidth of motion controller is low, thereby smoothening out such discontinuities

Toolbox

lspb(0,1,50)

lspb(0,-1,50)

s = lspb(0,1,50)

[s sd sdd] = lspb(0,1,50)

lspb(0,1,50, 0.025)
lspb(0,1,50, 0.035)

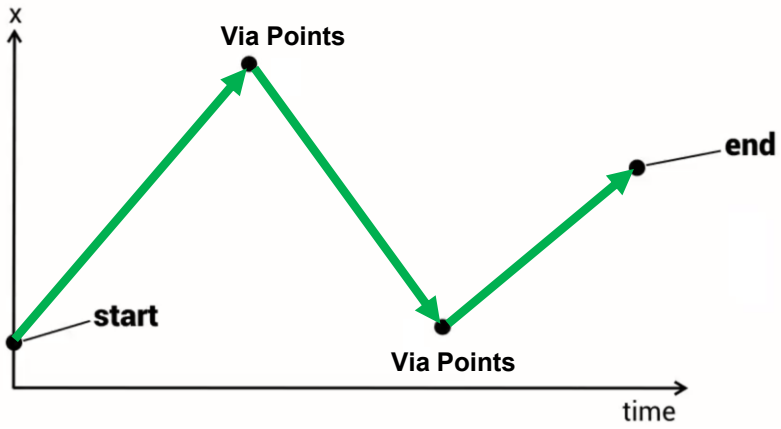
Multi-Segment Trajectories

- **Motivation**: Robots are required to smoothly move along a path with intermediate point (via points). This is important in many scenarios where obstacle avoidance is required or task requirement such as in welding or other manufacturing operations.

Formally:
• Consider that the trajectory is defined by M configurations q_k, $k \in [1, M]$ and there are M-1 motion segments. $q_k \in R^N$ is a *vector* representation of configuration.
• The robot moves from q_1 to q_m while moving through each of the intermediate points q_m. This leads to over constraint problem formulation
• To ensure continuity in velocity, we relax these constraints and skip some of the intermediate points and employ Blends (combination of linear and parabolic segments)

43

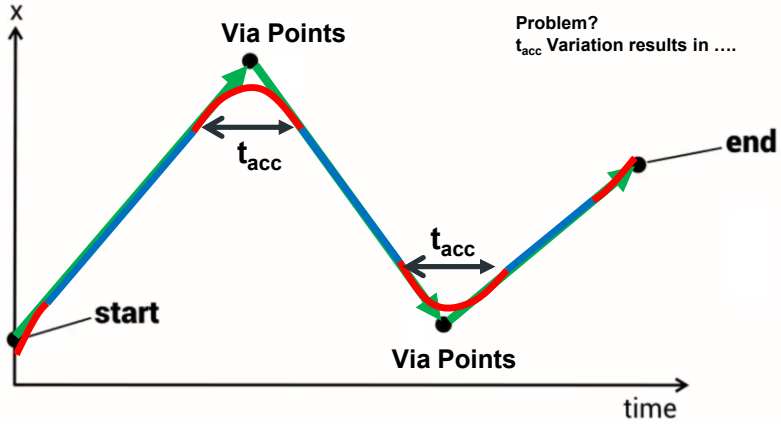
1D Trajectory with via points



Problem at via points? Discontinuity/abrupt changes …….not smooth trajectory

44

1D Trajectory with via points



Solution: Blends -> Smoothen out the curve (position, velocity and acceleration)
t_{acc} is the bend time
Path has multiple segments: Linear velocity + higher order polynomial blend

Multi-Segment Trajectories

- **Fundamental Engineering trade off between accuracy and acceleration**
- **If Tacc is small**
- **We go close to the via points, but acceleration is high.**
- **If Tacc is large**
- **Acceleration is low, but further from the via points.**
- **Elaborated in the graphs coming in next slides.**

Toolbox

```
first=10;
last=30;
via=[40; 10; last]

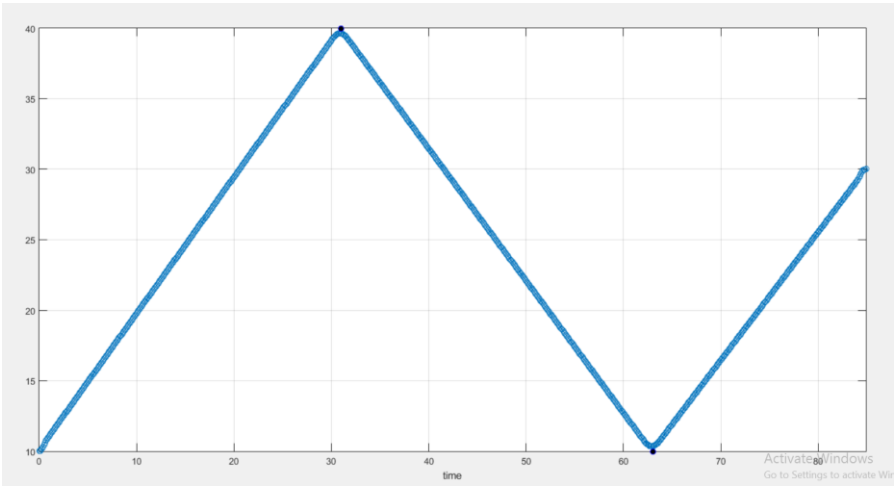
mstraj(via, 1, [], first, 0.1, 2)

%(segments, velocity, tsegment, q, dt, Tacc)
```

mstraj**(**segments**,** velocity, time segment, start location, time interval, acc time)

47

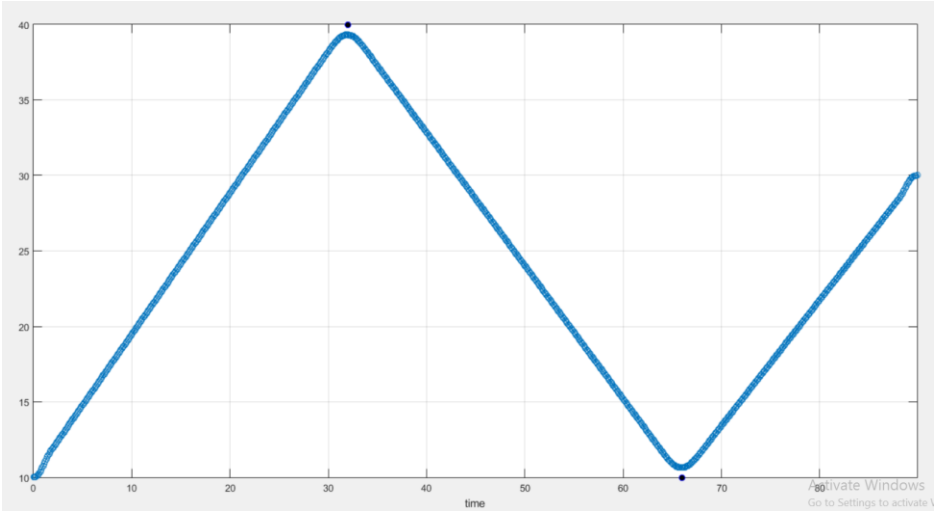
Toolbox



Tacc = 2

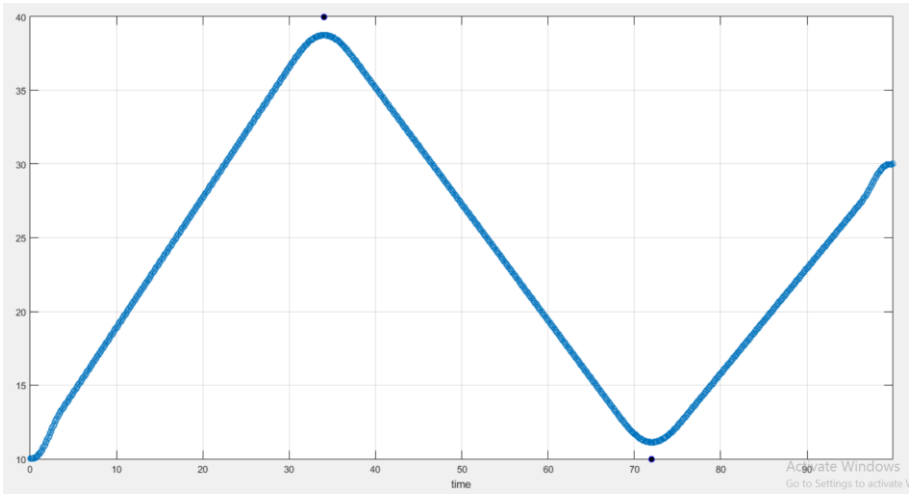
48

Toolbox



Tacc = 4

Toolbox



Tacc = 8

Toolbox

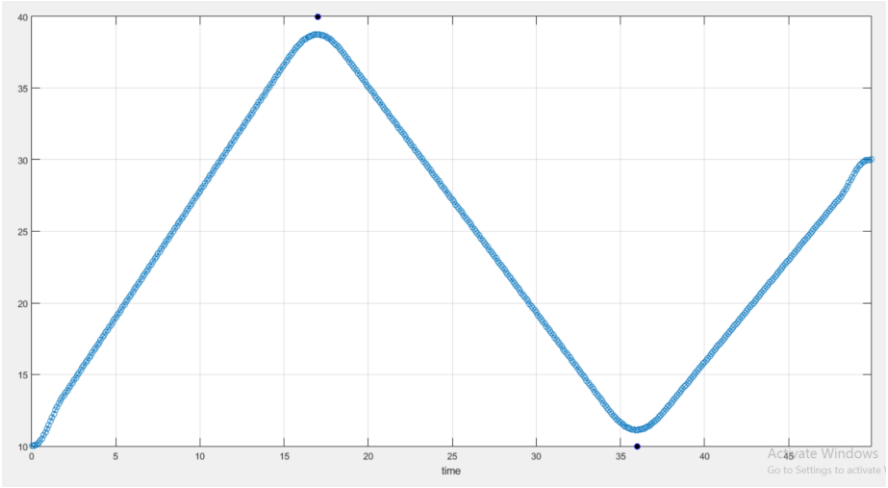
```
first=10;
last=30;
via=[40; 10; last]

mstraj(via, 2, [], first, 0.1, 4)

%Function Args
%(segments, velocity, tsegment, q, dt, Tacc)
mstraj(via, velocity, time, first location, time interval, acc time)
```

51

Toolbox



Velocity doubled to 2, Tacc = 4

52

Toolbox

```
first=10;
last=30;
via=[40; 10; last]

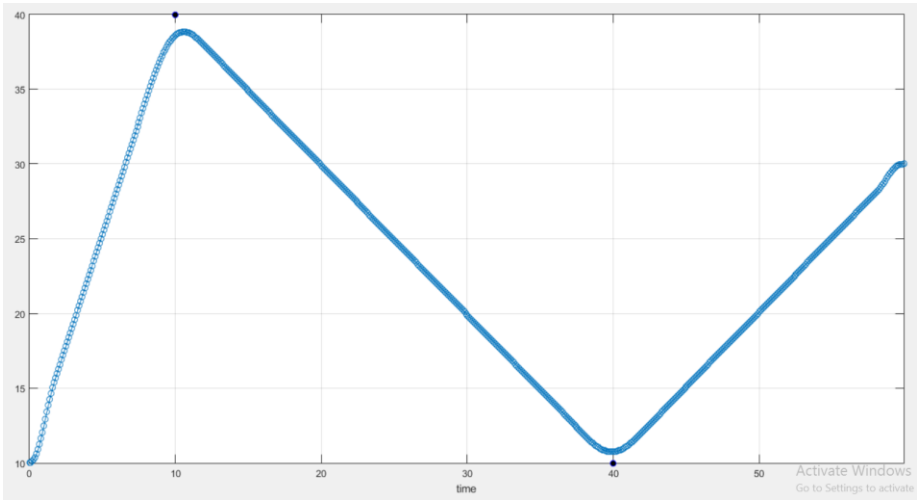
mstraj(via,[],[10 30 20], first, 0.1, 4) %(segments, velocity, tsegment, q, dt, Tacc)
```



Time for each segment

53

Toolbox



We are concerned about time for each segment rather than velocity

54

Task/Operational Space

- We consider the pose of the objects in terms of position and orientation of a coordinate frame affixed to them

- **Arm-Type Robot**: Affix a coordinate frame to its end-effector. Calculate all poses from base till the end-effector and save them.
- **Mobile Robot**: Affix a frame to its body (Body-fixed frame).
- This coordinate frame is sufficient to describe the *task space/operational space* of the robot in 2D/3D Euclidean space.

55

Configuration Space

- **Configuration** is the smallest set of parameters, called generalized coordinates, that are required to fully describe the position of *every* particle in the system.
- **In Mobile Robot**: All particles comprising the robot move together and a single generalized coordinate q is required to describe the location of all points.
- **In Robot Arm**: A robot arm with fixed-base and rigid links connected through rotational joints has a configuration that can be completely described by generalized coordinates – the joint angles (q_1, $q_{2....}$)

56

Credits

- This template is customized from a Slidesgo template.
- This lecture is made from the book “RVC by Professor Peter Corke”
- Some slides are inspired by Professor Peter Corke’s work.
- Rest of the material is self made and some part is compiled from internet resources.

slidesgo

Robotics – Umarah Qaseem

95